Tutorial 2

Linear mapping

$\mathbb{R}^n \rightarrow \mathbb{R}^m$

$$f(X_1, ..., X_n) \rightarrow \begin{bmatrix} a_{1,1}X_1 + ... + a_{1,n}X_n \\ a_{2,1}X_1 + ... + a_{2,n}X_n, \\ ... \\ a_{m,1}X_1 + ... + a_{m,n}X_n \end{bmatrix}$$

$$f((X_1, ..., X_n) \rightarrow Tx$$

$$T = \begin{bmatrix} a_{1,1}, a_{1,2}, ..., a_{1,n} \\ a_{2,1}, a_{2,2}, ..., a_{2,n} \\ ... \\ a_{m,1}, a_{m,2}, ..., a_{m,n} \end{bmatrix}$$

Valid linear transformation

let
T be the transformation matrix
f is the tranformation function $\mathbb{R}^n \rightarrow \mathbb{R}^m$
$x, y \in \mathbb{R}^n$
$c \in \mathbb{R}$, then

1. $f(0) \rightarrow T.0 \rightarrow 0$
2. $f(x + y) \rightarrow f(x) + f(y) \rightarrow T(x) + T(y)$
3. $f(cx) \rightarrow cf(x) \rightarrow cT(x)$

Question 1: Find which of the following transformations are valid linear transformations

1. $T : \mathbb{R}^2 \to \mathbb{R}^3, f(x, y) = (x, x + y, x + 1)$
2. $T : \mathbb{R}^2 \to \mathbb{R}^2, f(x, y) = (xy, x + y)$
3. $T : \mathbb{R}^2 \to \mathbb{R}^3, f(x, y) = (x + 2y, x + y + 1, 2x - y)$
4. $T : \mathbb{R}^3 \to \mathbb{R}^2, f(x, y, z) = (x + y - z, 2x)$
5. $T : \mathbb{R}^3 \to \mathbb{R}^3, f(x, y, z) = (\frac{x}{2}, \frac{y}{2}, \frac{z}{2})$
6. $T : \mathbb{R}^2 \to \mathbb{R}^2, f(x, y) = (\frac{x}{2}, \frac{xy}{2})$

Question 2: Find the transformation matrix $T : U \to V = T : \mathbb{R}^2 \to \mathbb{R}^2$

$$f(x_1, x_2) = \begin{bmatrix} x_1 - x_2 \\ 2x_1 + 3x_2 \end{bmatrix}$$

1. Use the standard basis for both vector space
2. Use the standard basis for U and $\{[1, 1], [-1, 1]\}$ for V
3. Use $\{[1, 1], [-1, 1]\}$ for U and $\{[1, 1], [-1, 2]\}$ for V

Solution

1.

$$f\begin{pmatrix}1\\0\end{pmatrix} = \begin{pmatrix}1\\2\end{pmatrix} = 1\begin{pmatrix}1\\0\end{pmatrix} + 2\begin{pmatrix}0\\1\end{pmatrix}$$

$$f\begin{pmatrix}0\\1\end{pmatrix} = \begin{pmatrix}-1\\3\end{pmatrix} = -1\begin{pmatrix}1\\0\end{pmatrix} + 3\begin{pmatrix}0\\1\end{pmatrix}$$

$$T = \begin{bmatrix}1, -1\\2, 3\end{bmatrix}$$

2.

$$f\begin{pmatrix}1\\0\end{pmatrix} = \begin{pmatrix}1\\2\end{pmatrix} = x\begin{pmatrix}1\\1\end{pmatrix} + y\begin{pmatrix}-1\\1\end{pmatrix}$$

Solve for x_1 and x_2 using system of linear equation method $Ax = b$

$$A = \begin{bmatrix}1, -1\\1, 1\end{bmatrix}, x = \begin{bmatrix}x_1\\x_2\end{bmatrix}, b = \begin{bmatrix}1\\2\end{bmatrix}$$

$$f\begin{pmatrix}1\\0\end{pmatrix} = \begin{pmatrix}1\\2\end{pmatrix} = \tfrac{3}{2}\begin{pmatrix}1\\1\end{pmatrix} + \tfrac{1}{2}\begin{pmatrix}-1\\1\end{pmatrix}$$

$$f\begin{pmatrix}0\\1\end{pmatrix} = \begin{pmatrix}-1\\3\end{pmatrix} = 1\begin{pmatrix}1\\1\end{pmatrix} + 2\begin{pmatrix}-1\\1\end{pmatrix}$$

$$T = \begin{bmatrix}\frac{3}{2}, 1\\\frac{1}{2}, 2\end{bmatrix}$$

3.

$$f\begin{pmatrix}1\\1\end{pmatrix} = \begin{pmatrix}0\\5\end{pmatrix} = \tfrac{5}{3}\begin{pmatrix}1\\1\end{pmatrix} + \tfrac{5}{3}\begin{pmatrix}-1\\2\end{pmatrix}$$

$$f\begin{pmatrix}-1\\1\end{pmatrix} = \begin{pmatrix}-2\\1\end{pmatrix} = -1\begin{pmatrix}1\\1\end{pmatrix} + 1\begin{pmatrix}-1\\2\end{pmatrix}$$

$$T = \begin{bmatrix}\frac{5}{3}, -1\\\frac{5}{3}, 1\end{bmatrix}$$

Question 3: Find the transformation matrix

$T : U \to V = T : \mathbb{R}^2 \to \mathbb{R}^3$

$$f(x_1, x_2) = \begin{bmatrix} x_1 + 3x_2 \\ 4x_1 - x_2 \\ 2x_1 + x_2 \end{bmatrix}$$

1. Use standard basis for both U and V

2. Use the following basis $\{[1, 1], [-1, 2]\}$ for U and $\{[1, 1, 0], [0, 1, 1], [1, 0, 1]\}$ for V

Solution

1.

$$f\begin{pmatrix}1\\0\end{pmatrix} = \begin{pmatrix}1\\4\\2\end{pmatrix} = 1\begin{pmatrix}1\\0\\0\end{pmatrix} + 4\begin{pmatrix}0\\1\\0\end{pmatrix} + 2\begin{pmatrix}0\\0\\1\end{pmatrix}$$

$$f\begin{pmatrix}0\\1\end{pmatrix} = \begin{pmatrix}3\\-1\\1\end{pmatrix} = 3\begin{pmatrix}1\\0\\0\end{pmatrix} + -1\begin{pmatrix}0\\1\\0\end{pmatrix} + 1\begin{pmatrix}0\\0\\1\end{pmatrix} \quad T = \begin{bmatrix}1,3\\4,-1\\2,1\end{bmatrix}$$

2.

$$f\begin{pmatrix}1\\1\end{pmatrix} = \begin{pmatrix}4\\3\\3\end{pmatrix} = 2\begin{pmatrix}1\\1\\0\end{pmatrix} - 1\begin{pmatrix}0\\1\\1\end{pmatrix} + 2\begin{pmatrix}1\\0\\1\end{pmatrix}$$

$$f\begin{pmatrix}-1\\2\end{pmatrix} = \begin{pmatrix}5\\-6\\0\end{pmatrix} = -\frac{1}{2}\begin{pmatrix}1\\1\\0\end{pmatrix} - \frac{11}{2}\begin{pmatrix}0\\1\\1\end{pmatrix} + \frac{11}{2}\begin{pmatrix}1\\0\\1\end{pmatrix}$$

$$T = \begin{bmatrix}2,-\frac{1}{2}\\1,-\frac{11}{2}\\2,\frac{11}{2}\end{bmatrix}$$

Four fundamental subspaces

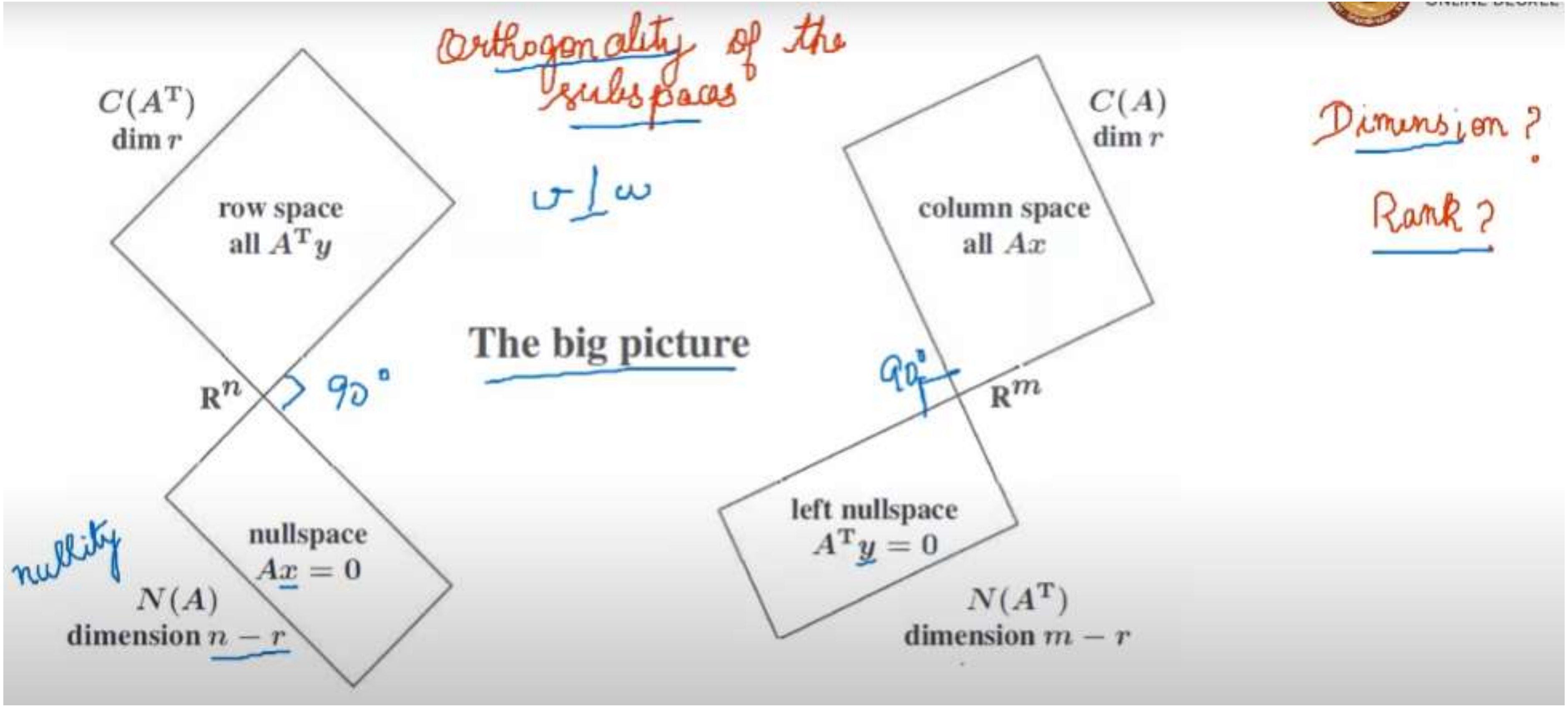
Suppose $\mathbf{A}$ is a $m \times n$ matrix.

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

1. The column space is $C(\mathbf{A})$, a subspace of $\mathbb{R}^m$.
2. The row space is $C(\mathbf{A}^T)$, a subspace of $\mathbb{R}^n$.
3. The nullspace is $N(\mathbf{A})$, a subspace of $\mathbb{R}^n$.
4. The left nullspace is $N(\mathbf{A}^T)$, a subspace of $\mathbb{R}^m$.

Source: https://www.youtube.com/watch?v=rbBiDiFC584

Orthogonality of the subspaces



Question 4: Obtain the four fundamental spaces of A and find its rank and nullity

$$A = \begin{bmatrix} 2,4,6,8,10 \\ 2,0,2,4,-2 \\ 2,2,4,0,2 \\ 4,4,8,12,8 \end{bmatrix}$$

1. find the column space
2. find the null space
3. find the row space
4. find the left null space

Column space

Gaussian elimination

Row echelon form

pivot

$$\begin{bmatrix} \boxed{2} & 4 & 6 & 8 & 10 \\ 2 & 0 & 2 & 4 & -2 \\ 2 & 2 & 4 & 0 & 2 \\ 4 & 4 & 8 & 12 & 8 \end{bmatrix}$$

$R_2 - R_1$

$R_3 - R_1$

$R_4 - 2R_1$

$$\begin{pmatrix} 2 & 4 & 6 & 8 & 10 \\ 0 & -4 & -4 & -4 & -12 \\ 0 & -2 & -2 & -8 & -8 \\ 0 & -4 & -4 & -4 & -12 \end{pmatrix}$$

$R_3 - \frac{1}{2}R_2$

$R_4 - R_2$

pivots

$$\begin{pmatrix} \boxed{2} & 4 & 6 & 8 & 10 \\ 0 & \boxed{-4} & -4 & -4 & -12 \\ 0 & 0 & 0 & \boxed{-6} & -2 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

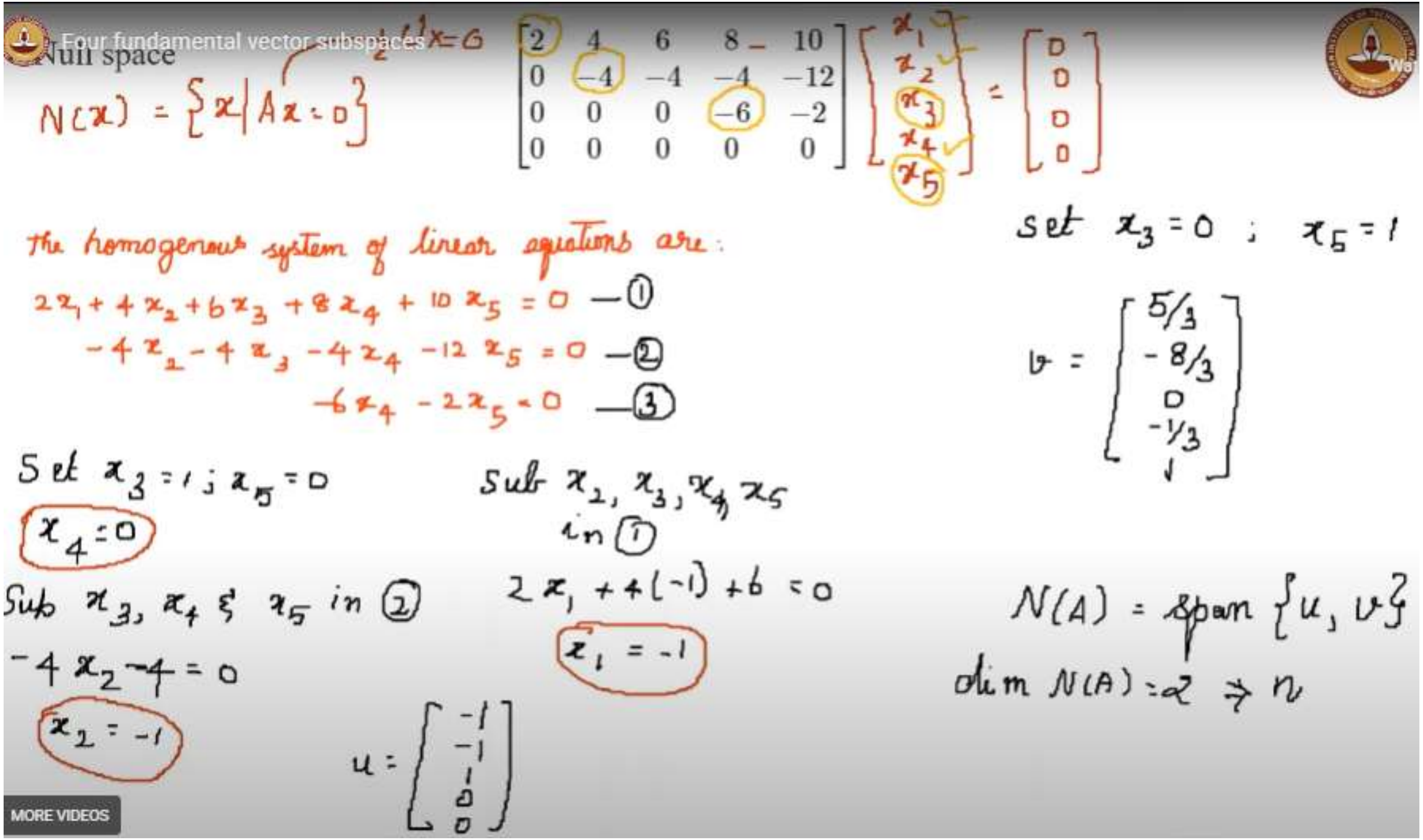
Row echelon form

column space

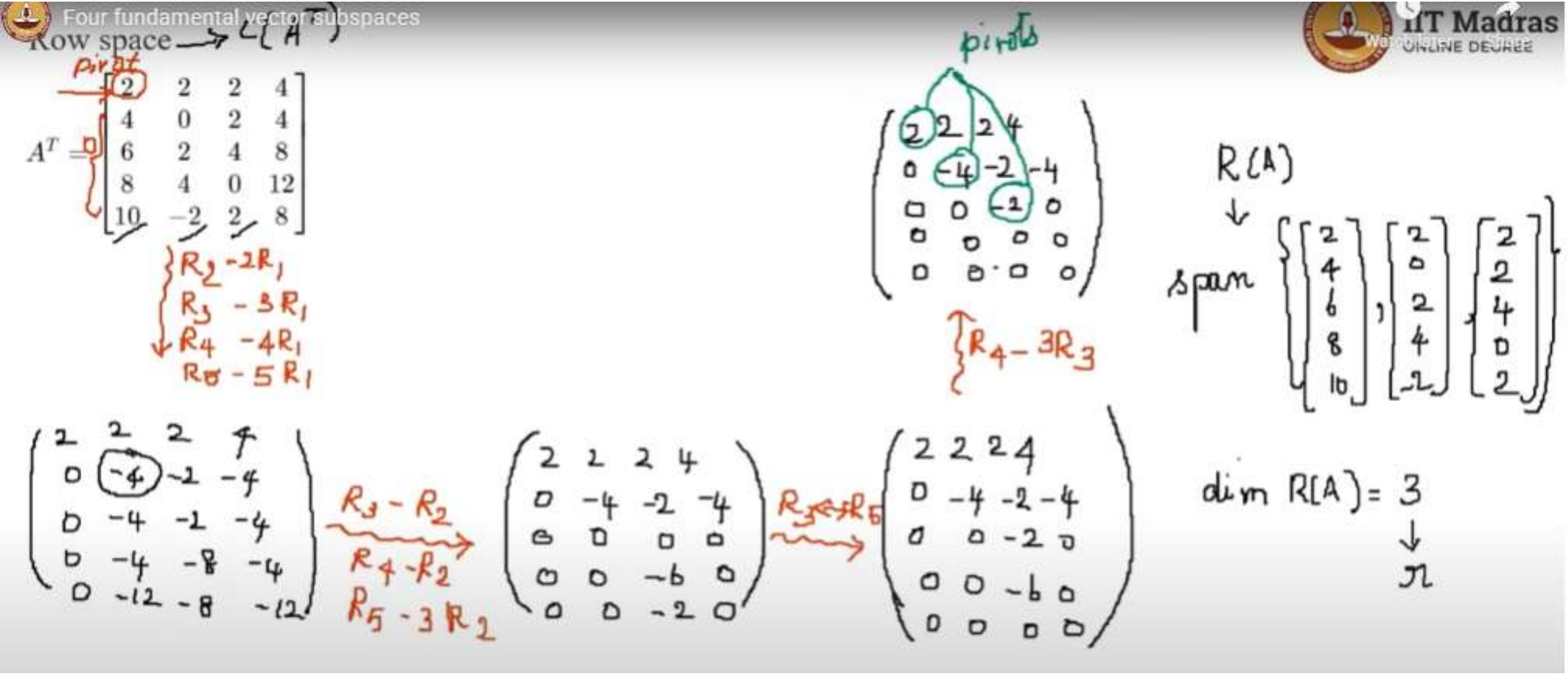
$$C(A) = \text{span}\left\{ \begin{bmatrix} 2 \\ 2 \\ 2 \\ 4 \end{bmatrix}, \begin{bmatrix} 4 \\ 0 \\ 2 \\ 4 \end{bmatrix}, \begin{bmatrix} 8 \\ 4 \\ 0 \\ 12 \end{bmatrix} \right\}$$

$\text{rank} = 3 \Rightarrow \dim C(A)$

Nullspace


Four fundamental vector subspaces
Null space
$N(x) = \{x \mid Ax = 0\}$
$\begin{bmatrix} 2 & 4 & 6 & 8 & 10 \\ 0 & -4 & -4 & -4 & -12 \\ 0 & 0 & 0 & -6 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$
the homogenous system of linear equations are:
$2x_1 + 4x_2 + 6x_3 + 8x_4 + 10x_5 = 0$ — (1)
$-4x_2 - 4x_3 - 4x_4 - 12x_5 = 0$ — (2)
$-6x_4 - 2x_5 = 0$ — (3)
Set $x_3 = 1$; $x_5 = 0$
$x_4 = 0$
Sub x_3, x_4 & x_5 in (2)
$-4x_2 - 4 = 0$
$x_2 = -1$
Sub x_2, x_3, x_4, x_5 in (1)
$2x_1 + 4(-1) + 6 = 0$
$x_1 = -1$
$u = \begin{bmatrix} -1 \\ -1 \\ 1 \\ 0 \\ 0 \end{bmatrix}$
set $x_3 = 0$; $x_5 = 1$
$v = \begin{bmatrix} 5/3 \\ -8/3 \\ 0 \\ -1/3 \\ 1 \end{bmatrix}$
$N(A) = \text{span}\{u, v\}$
$\dim N(A) = 2 \Rightarrow n$
MORE VIDEOS

Row space


Four fundamental vector subspaces
Row space → C(A^T)
IIT Madras
ONLINE DEGREE
pivot
pivots
R2 - 2R1
R3 - 3R1
R4 - 4R1
R5 - 5R1
R3 - R2
R4 - R2
R5 - 3R2
R3 ↔ R5
R4 - 3R3
R(A)
span
dim R(A) = 3

Left null space